

HaptiX: A Fusion Framework for Spatiotemporal Cross-Modal Haptic Synthesis on Mobile Devices

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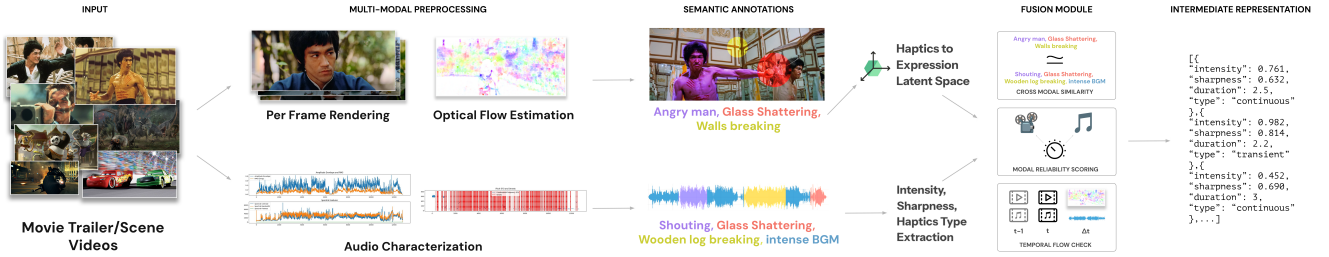
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Graphical Abstract: Cross-Modal Haptic Synthesis at a Glance

Abstract

The rise of mobile entertainment platforms has significantly transformed global media consumption habits. This work presents Haptix - a comprehensive framework for generating context-aware haptics tailored to cinematic video inputs on mobile devices. The primary objective is to produce spatiotemporally aligned haptic responses that enhance user immersion and engagement. Evaluation through user studies reveals strong support for the system's effectiveness. While both novice and experienced users reported high levels of synchronization and experiential enhancement, experienced users demonstrated significantly greater comfort, reduced fatigue, and a higher intention to adopt the technology. These findings underscore the value of adaptive system design with novice users benefiting from on-boarding and calibration mechanisms, and experienced users expecting high-fidelity, tightly synchronized

feedback. The results point toward a promising future for haptic media, where user familiarity and context-aware design will play central roles in delivering immersive, emotionally resonant experiences, thus enriching digital storytelling.

Keywords

Haptics Generation, Immersive Multimedia, Scene Understanding, Human Computer Interaction

ACM Reference Format:

Veeramanohar Avudaiappan, Sakthivel Saravanathangam, Nithiesh P Natarajan, and Ritwik Murali. 2025. HaptiX: A Fusion Framework for Spatiotemporal Cross-Modal Haptic Synthesis on Mobile Devices. In *SIGGRAPH Asia 2025 Posters (SA Posters '25)*, December 15–18, 2025, Hong Kong, Hong Kong. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3757374.3771521>

1 Introduction

The rapid rise of mobile entertainment—from OTT streaming to short-form apps like TikTok and YouTube Shorts—has reshaped global media consumption. Despite the ubiquity and multimodal potential of mobile devices, content remains largely audiovisual, overlooking the tactile dimension that could enrich immersion.

Research shows that synchronizing visual, auditory, and haptic cues significantly improves realism and emotional engagement, but most apps prioritize visuals, leading to poor cross-modal timing and reduced user experience [Zhao and Cen 2024]. To bridge

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SA Posters '25, Hong Kong, Hong Kong

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ACM ISBN 979-8-4007-2134-2/25/12

<https://doi.org/10.1145/3757374.3771521>

this gap, we introduce a platform-agnostic intermediate representation framework that enables real-time, hardware-independent haptic integration across iOS and Android. By supporting automated spatiotemporal synchronization and providing user-level creative control, our approach facilitates accessible and immersive multisensory media experiences on mobile platforms.

2 Methodology

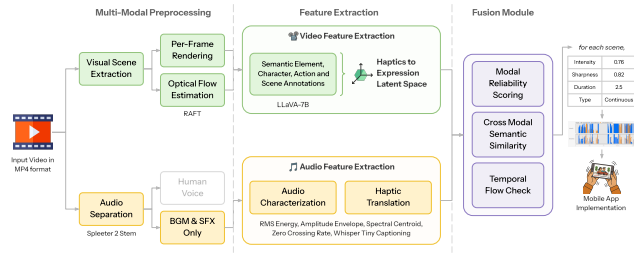


Figure 1: HaptiX Framework - Workflow

As shown in figure 1, the workflow starts with an MP4 video input processed for both audio and video analysis. In the *Scene Segmentation* stage, the video is divided into scenes by first scoring frames based on color differences, and then applying a rolling average to refine cut points, accurately detecting fast cuts. *Frame Rendering and Motion Analysis* extracts key frames from each scene and computes optical flow using the Recurrent All-Pairs Field Transforms (RAFT) deep neural network to capture object movement, character actions, and environmental changes. Similarly, for *audio extraction & processing*, Spleeter’s two-stem model is used to decompose audio into human voices, and a combined track of background music (BGM) and sound effects (SFX). The human voices (dialogues) are then excluded to prevent haptic responses to speech. The resulting BGM+SFX track is then denoised and enhanced to improve clarity and support more effective feature extraction.

In the *feature extraction* stage, both visual and auditory streams are analyzed to produce semantically rich inputs for haptic translation. For *visual features*, each video frame is processed through LLaVA-7B to generate descriptive annotations of key scene elements, including characters, actions, motion cues, and environmental context [Murali and Avudaiappan 2024; S and Namitha 2025]. These structured annotations serve as intermediaries that map visual semantics into meaningful haptic patterns.

The extracted haptic features are mapped to semantic labels representing emotional states, contextual cues, and intended effects. A *haptic latent embedding space*, derived from the open-source VibViz dataset [Seifi 2016], facilitates alignment between visual annotations and relevant haptic patterns. For the audio stream, the denoised BGM and SFX tracks are analyzed to compute descriptors such as mean-square energy, spectral centroid, zero-crossing rate, and spectral flux—capturing attributes like loudness, pitch, and temporal dynamics. Complementing these low-level features, *audio captioning* using the Whisper Tiny model generates high-level descriptions (e.g., “dramatic music,” “sudden crash”), enriching contextual cues for haptic response generation.

After independently extracting video and audio features, the *Fusion Module* generates spatiotemporally aligned haptic responses via a three-step process. Dynamic *modal reliability scoring* assigns weights based on each modality’s relevance, *cross-modal semantic similarity* compares video and audio to prioritize the more contextually relevant input, and a *temporal flow check* smooths transitions by ensuring continuity with previous haptic outputs.

3 Implementation & User Feedback

The fused haptic data is first converted into a platform-agnostic format for compatibility, and then translated into device-specific outputs like AHAP for iOS and vibration patterns for Android. A custom script integrated within the mobile framework automates this conversion, enabling seamless cross-platform deployment. The effectiveness of the approach is validated through user studies involving 50 participants, evaluating immersion, synchronization, and overall user comfort. Group A (Users 1–25) had no prior haptic experience, while Group B (Users 26–50) were familiar with haptics from gaming. Participants rated the system on seven dimensions, namely, synchronization, semantic reflection, experiential enhancement, engagement, comfort, fatigue, and future adoption. Group A’s average scores ranged from 6.69 to 6.96 across most metrics, with 2.88 fatigue and 80% interested in future adoption. Group B showed higher ratings (8.04 to 7.81) with lower fatigue (1.96) and 92% adoption interest. Comparative analysis shows prior exposure enhances the haptic experience. Both groups scored above 6/10 in key areas, showing the system’s effectiveness. Results confirmed that familiarity improves satisfaction, lowers cognitive load, and increases adoption. This highlights the need for adaptive designs, onboarding for novices, high-fidelity feedback for experts, and supports the pipeline’s robustness and potential for entertainment integration.

4 Conclusion & Future Directions

This work introduces a context-aware framework for generating spatiotemporally aligned haptic feedback from cinematic video content on mobile devices. The three-phase pipeline—comprising multi-modal preprocessing, feature extraction, and haptic fusion—translates audio-visual input into semantically meaningful tactile responses. User studies validate the system’s effectiveness, with users reporting higher comfort, reduced fatigue, and stronger adoption intent. Overall, the results demonstrate the potential of haptic media to enrich digital storytelling through immersive and emotionally engaging experiences. Future directions could explore personalized haptic profiling, real-time processing, and even scalable integration of context-aware haptics into multimodal media systems.

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